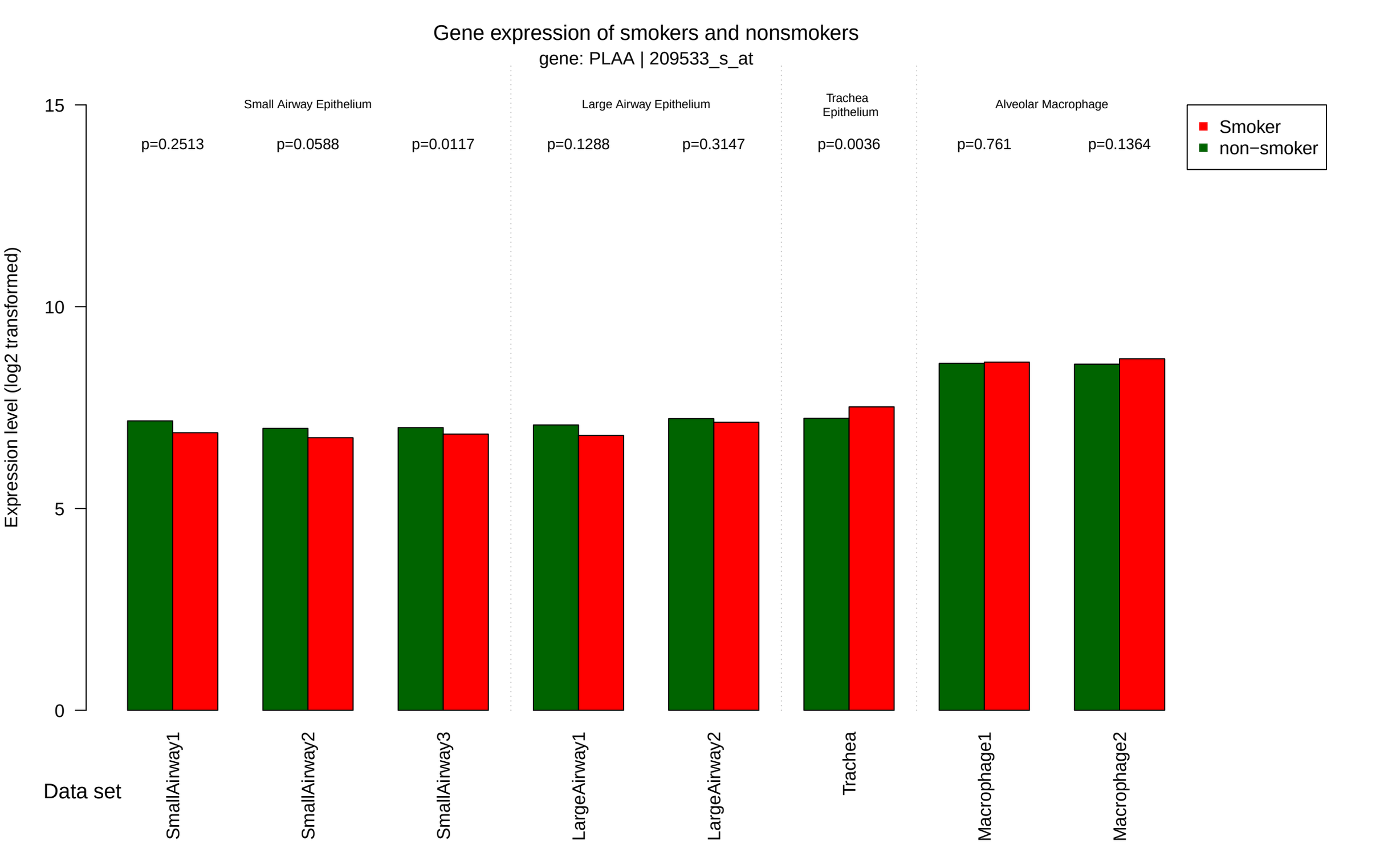
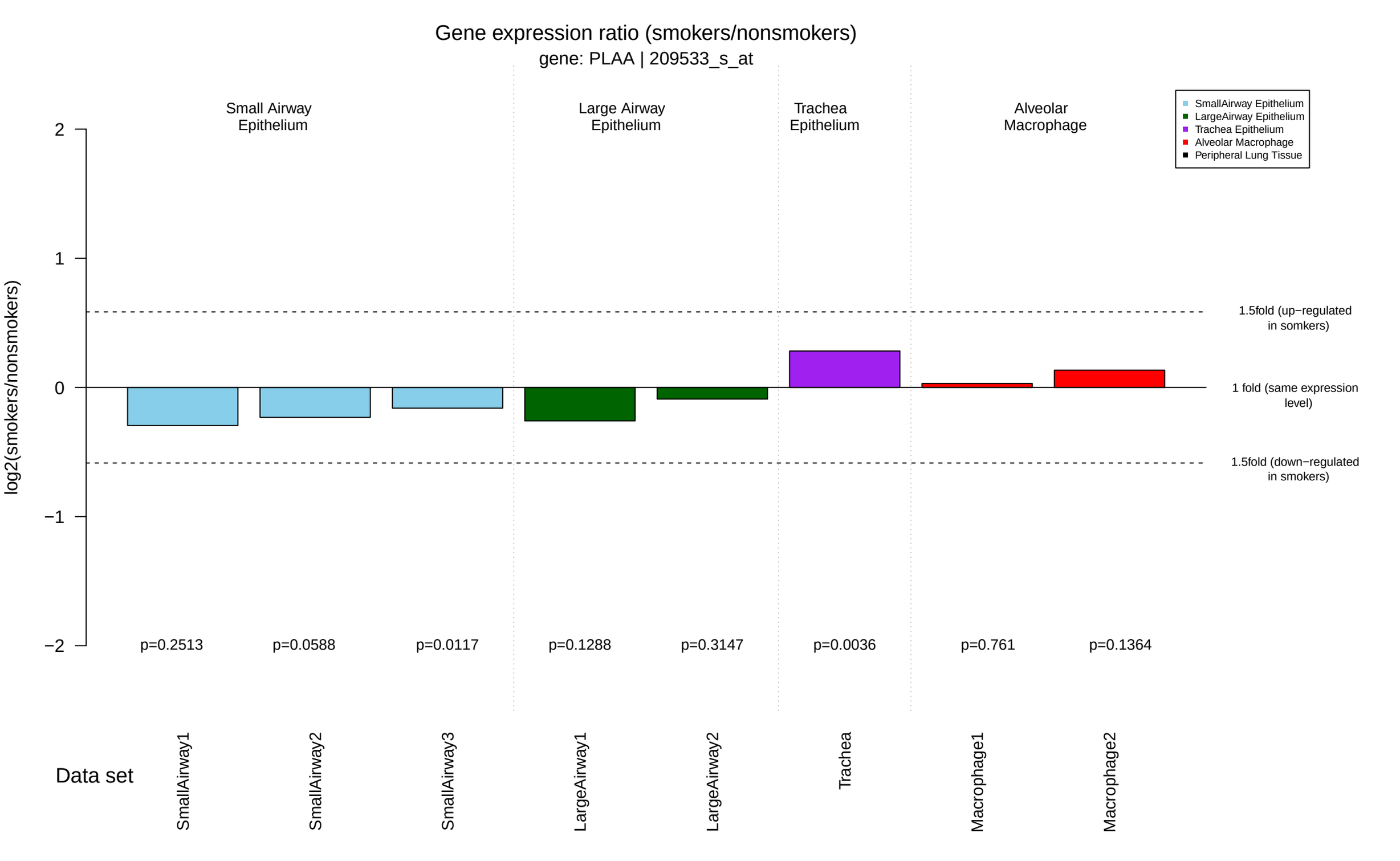
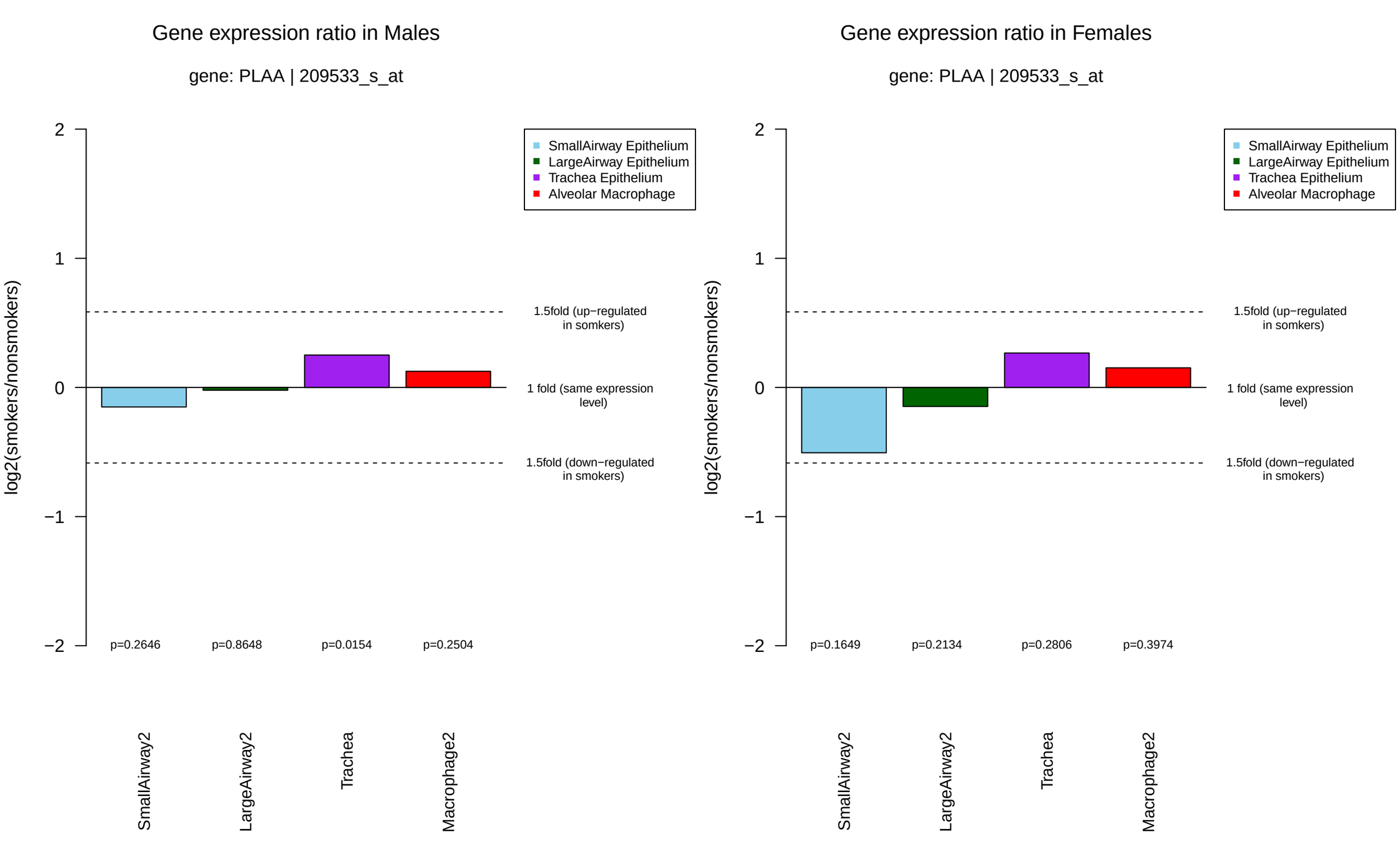
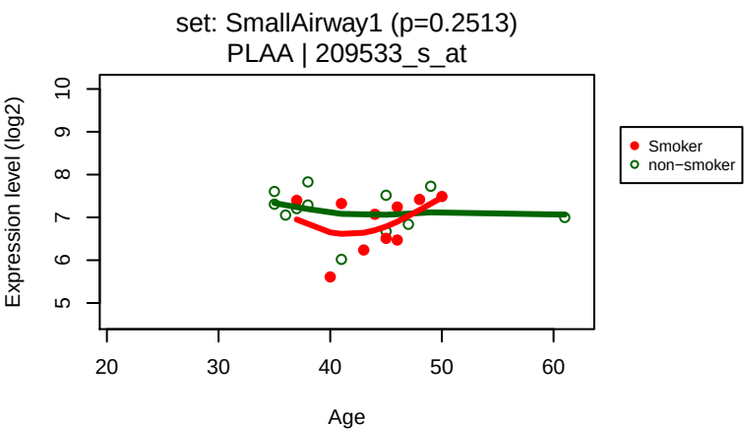
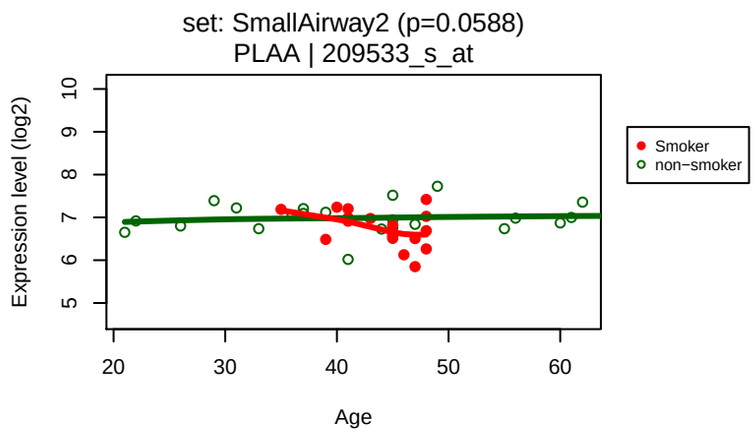
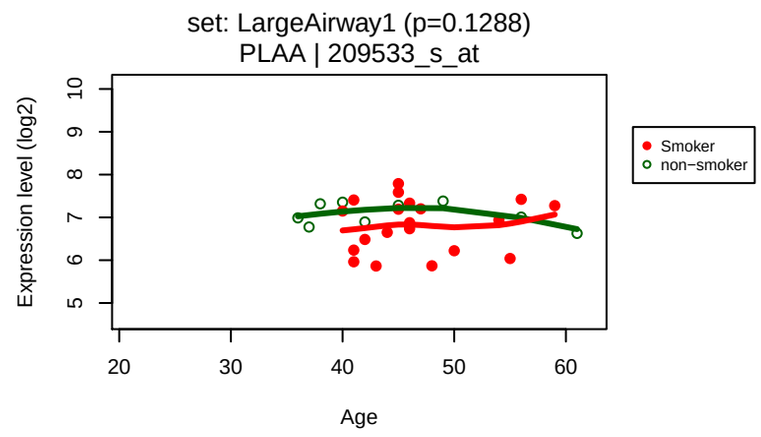

Gene expression of smokers and nonsmokers
gene: PLAA | 209533_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2513
p=0.0588
p=0.0117
p=0.1288
p=0.3147
p=0.0036
p=0.761
p=0.1364
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

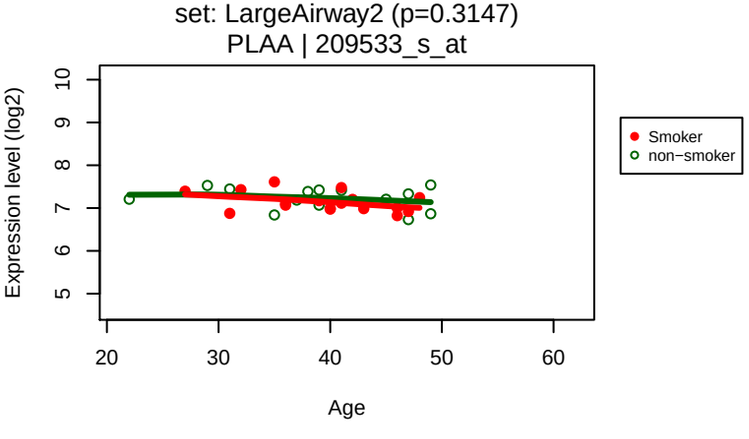

Gene expression ratio (smokers/nonsmokers)
gene: PLAA | 209533_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2513
p=0.0588
p=0.0117
p=0.1288
p=0.3147
p=0.0036
p=0.761
p=0.1364
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

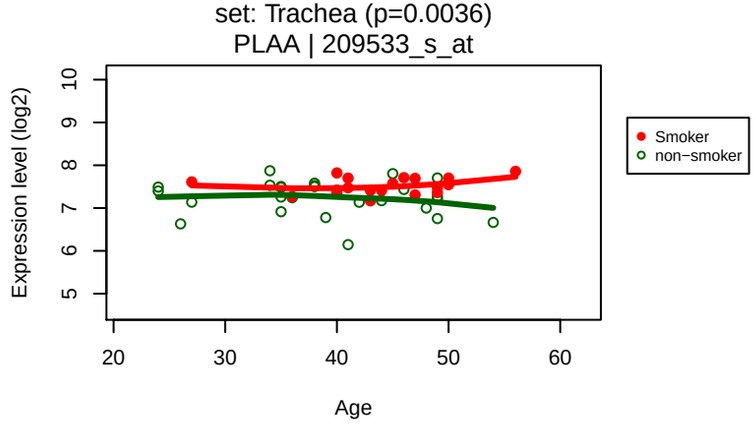

Gene expression ratio in Males
gene: PLAA | 209533_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2646
p=0.8648
p=0.0154
p=0.2504
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLAA | 209533_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1649
p=0.2134
p=0.2806
p=0.3974
SmallAirway2
LargeAirway2
Trachea
Macrophage2

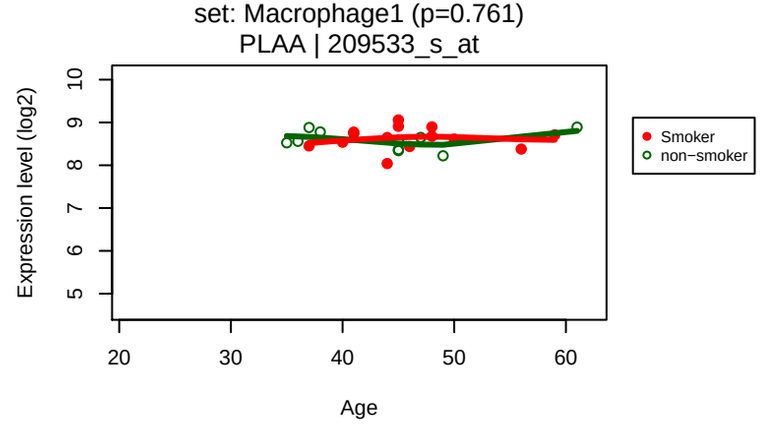

set: SmallAirway1 (p=0.2513)
PLAA | 209533_s_at
Expression level (log2)
Age
Smoker
non−smoker

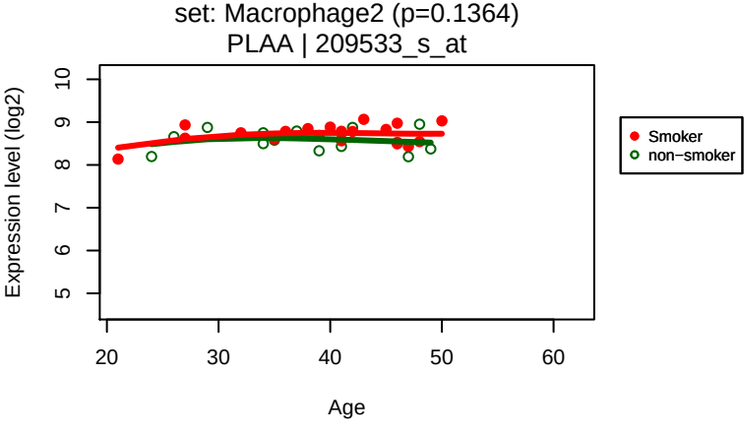

set: SmallAirway2 (p=0.0588)
PLAA | 209533_s_at
Expression level (log2)
Age
Smoker
non−smoker

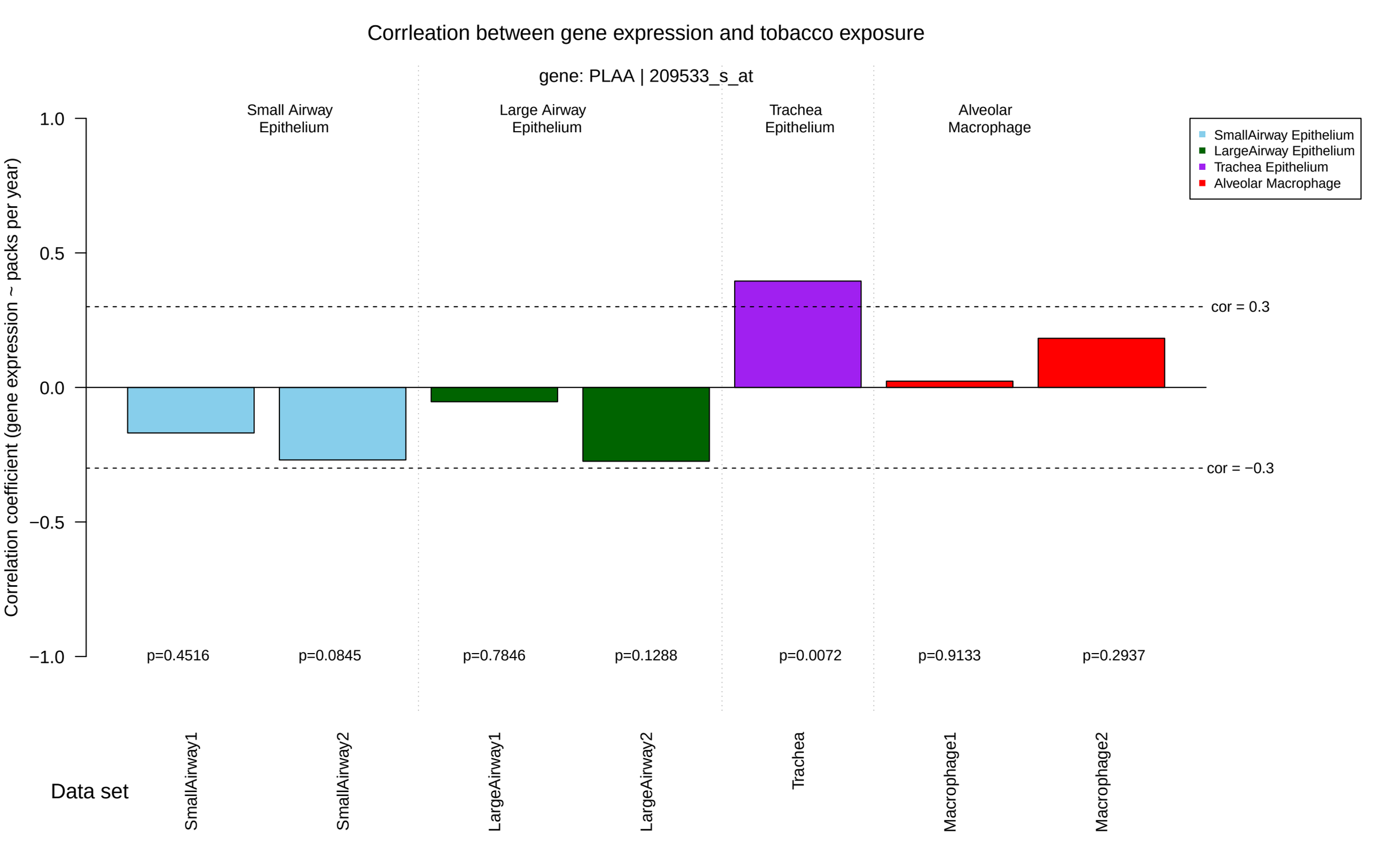

set: LargeAirway1 (p=0.1288)
PLAA | 209533_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3147)
PLAA | 209533_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0036)
PLAA | 209533_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.761)
PLAA | 209533_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1364)
PLAA | 209533_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLAA | 209533_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4516
p=0.0845
p=0.7846
p=0.1288
p=0.0072
p=0.9133
p=0.2937
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2